Programming Lab – Argumentation Technology

Assignment 1: Argument Acquisition

Carlotta Quensel

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April 30, 2025

Natural Language Processing

Semester Schedule

1. Warm-up	○	15/04	Introduction & organization	1101.F138
	○	22/04	Presentation (submission: 20/04, 23:59h)	1101.F138
2. Data ✔	•	29/04	Introduction	1101.F138
	○	06/05	Q&A	1101.F138
	•	13/05	Presentation (submission: 11/05, 23:59h)	**!**3408.010
3. Mining ✔	•	20/05	Introduction	**!**3408.010
		~~27/05~~	*Pentecost week* (Q&A via StudIP)	——
	•	03/06	Presentation (submission: 01/06, 23:59h)	1101.F138
4. Assessment ✔	•	10/06	Introduction	1101.F138
	○	17/06	Q&A	1101.F138
	•	24/06	Presentation (submission: 22/06, 23:59h)	1101.F138
5. Generation ✔	•	01/07	Introduction	1101.F138
	○	08/07	Q&A	1101.F138
	•	15/07	Presentation (submission: 13/07, 23:59h)	1101.F138

✔: graded assignment; •: mandatory attendance; **!**: sessions start at 15:30

Attendance Check

Attendance Note — For the mandatory sessions, one authorized absence is permitted (i.e., you must notify me by email with a valid justification).

Semester Schedule

Assignment
Overview
Motivation
Data Structure
Tasks
1 – Preprocessing
2 – Exploration
3 – Analysis
Grading

Assignment 2 – Argument Acquisition

Overview

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Learning Goals

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- Distinguish relevant information from API outputs

Overview

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Topics Covered

- Working with JSON list data mirroring the API output of Reddit
- Extracting relevant data from these structures
- Computing and plotting basic statistics
- Transforming and splitting data for training and evaluation

Deadline: Monday, May 12, 2025 at 23:59h

Motivation

To study arguments – we need arguments

- News comment sections (websites/social media)
- Dedicated discussion or debate forums
- Official deliberative democracy platforms
- Parliamentary proceedings (speeches, debates)
- Political talk shows

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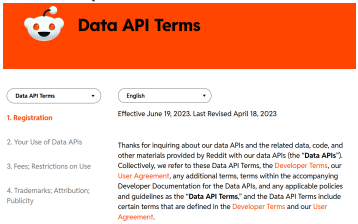
2. No luck? → Check usage restrictions (`robots.txt` or explicit guidelines)

```
User-agent:  Googlebot
Disallow:
User-agent:  Bingbot
Disallow:
User-agent:  *
Disallow:  /
```



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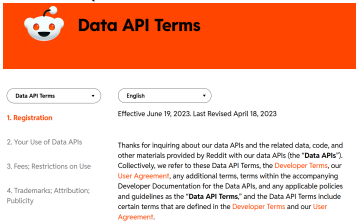

3. If unsure, ask (contact the rights holders via email, contact form) – never illicitly use data → legal/academic consequences

Application

Process discussions scraped from r/ChangeMyView by Tan et al. (2016) into a usable dataset for argument quality assessment.

Given an argument, predict its persuasiveness, i.e., if it was awarded a Δ.

Topic: *CMV: The position of Vice President of the United States should be eliminated from our government.*
Because of the fact that the Vice President is responsible for Presidential duties when the President is away means that they are the perfect person to take over for them. In the event of whatever emergency would surround the need for a VP to take the President's spot, the experience that the VP already has with their duties would mean they could easily step into the President's place without disrupting the rest of the cabinet, all of whom would likely be very busy with their current departments. $\rightarrow \Delta$

Data Structure

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You should therefore first load the data and acquaint yourself with its format.

- Visit the dataset's homepage to consult the Readme and Blog tutorial.
- Extract the data from `.data/cmv_reddit_api_data.jsonlist.bz2`.
- Explore the data: what does an instance look like, which information is encoded, and how are comment threads stored?

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Move on to the graded tasks once you feel confident with the dataset structure.

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Extract all posts from all original threads into one dataset and save it.

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- Add all comments to a `pandas.DataFrame` with column names *exactly* matching the above list of features (*-1 point per column/error*).
- Make sure that you only include posts that are usable for prediction, i.e., posts with unique, non-empty text fields (*3 points*).
- Save the data to `./outputs/posts.csv` without the index (*1 point*).

10 points

Task 2 – Data Exploration

- Now, we can analyze patterns of persuasive discussion and interaction
- For this, first calculate relevant metrics and plot them in a sensible way

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- Use `DataFrame.to_markdown()` to round floats to three decimal points; no formatting/rounding should be done beforehand.

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- Keep track of all other results (standard deviations, ❸).

Task 3 – Analysis

Given both subtasks from task 2, analyze and interpret your observations (*5 points each*).

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- **Evidence** – Include specific numbers. Copy the visualizations into your answer here (*1 point*).
- **Interpretation** – Reason about why this observation might be happening and what it may mean about persuasive behavior (*2 points*).

10 points

Grading

Point Requirements

- Points given for running code without preventable errors, following the task description in using/not using packages/methods as indicated
- Points given for correct outputs and correct full-text answers

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- Clear authorship violations (plagiarism, AI solutions) fail without grading

Bibliography

Chenhao Tan, Vlad Niculae, Cristian Danescu-Niculescu-Mizil, and Lillian Lee. 2016. Winning arguments: Interaction dynamics and persuasion strategies in good-faith online discussions. In *Proceedings of the 25th International Conference on World Wide Web*, WWW '16, pages 613–624, Republic and Canton of Geneva, CHE. International World Wide Web Conferences Steering Committee.